Click-to-Model: Real-Time Interactive Object Modeling and Robust 6D Pose Tracking

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Introduction

Robotic manipulation needs accurate 6D poses of unseen objects. Existing pipelines often require pre-scanned CAD, lack metric scale, or incur high tracking costs. CLM creates a metric object model from RGB-D and a user click, then combines learned localization with efficient geometric tracking.

Main contributions

Complete interactive perception pipeline

One click unifies object segmentation, 3D modeling, scale recovery, and online 6D pose tracking.

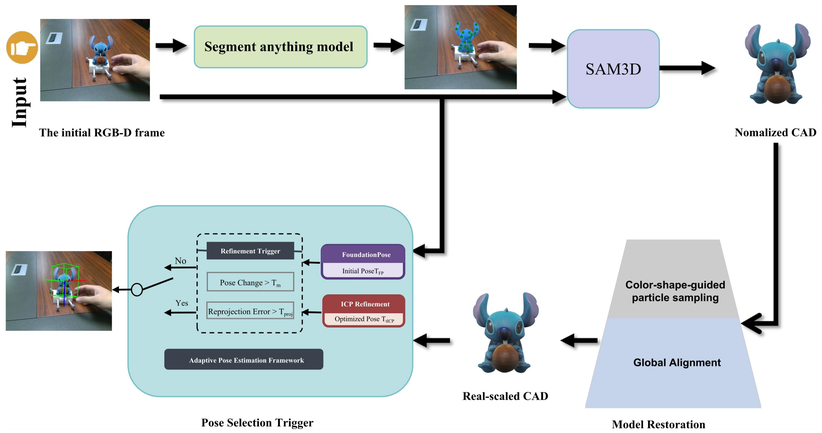
Scale-aware probabilistic registration

Joint color–shape cues select distinctive surface particles for robust metric alignment with RGB-D depth.

Decision-driven hybrid localization

A confidence-aware switch combines FoundationPose with ICP to balance tracking accuracy and runtime.

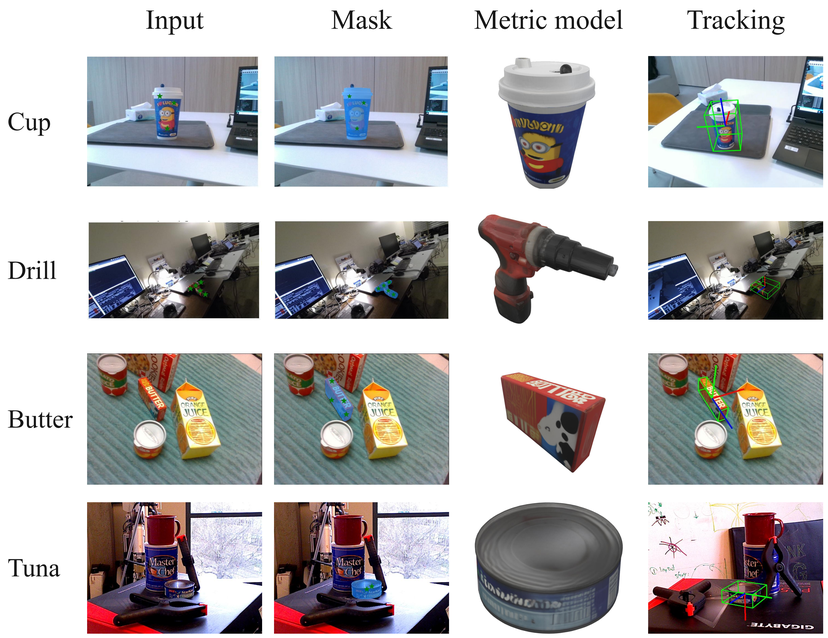
Pipeline framework



Online modeling requires no pre-scanned CAD. A new click reinitializes the tracker after target loss.

Qualitative tracking results

RGB-D input, interactive masks, metric-scale models, and pose tracking across diverse objects and scenes.



Examples from self-collected, HOPE, YCB-V, and NVIDIA scenes (paper, Fig. 3).

Experimental setup

NVIDIA RTX 4090 and Intel i9; Ubuntu 22.04 and Python. Aligned RGB-D images at 640 × 480, 30 FPS. Online tracking is measured after warm-up over 300 frames.

System-level comparison

Method	No CAD	General-ization	Inter-action	Real-time tracking	Metric scale
NOCS	No	No	No	Limited	Yes
FS-Net / SPD	No	No	No	Limited	Yes
ZeroPose	No	No	No	No	Partial
Any6D	Yes	Yes	No	No	Yes
FoundationPose	No	Yes	No	Yes	Yes
CLM (Ours)	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

Main formulas

Metric-scale recovery

Color–shape weights and depth anchors align the normalized model to physical dimensions.

$$s^* = \frac{\sum_{m=1}^{M} \omega_m \mathbf{a}_m^\top \mathbf{R}^\top (\mathbf{b}_m - \mathbf{t})}{\sum_{m=1}^{M} \omega_m \|\mathbf{a}_m\|_2^2}$$

Adaptive pose selection

Motion, depth error, and confidence select global estimation or efficient local refinement.

$$\alpha_t = \mathbb{I}\big(\lambda_1 \delta_t + \lambda_2 E_{depth}(\mathbf{T}_t^{FP}) + \lambda_3 (1 - c_t) > \tau\big)$$

$$\mathbf{T}_t^* = \begin{cases} \mathbf{T}_t^{FP}, & \alpha_t = 1, \\ \mathbf{T}_t^{ICP}, & \alpha_t = 0. \end{cases}$$

ICP-based local refinement

Small inter-frame motion permits point-to-plane alignment with the previous pose.

$$\Delta\mathbf{T}_t = \arg\min_{\Delta\mathbf{T}} \sum_i \Big(\mathbf{n}_{\pi(i)}^\top \big(\Delta\mathbf{T}\mathbf{T}_{t-1}^* \mathbf{s}_i - \mathbf{t}_{\pi(i)}\big)\Big)^2$$

$$\mathbf{T}_t^{ICP} = \Delta\mathbf{T}_t \mathbf{T}_{t-1}^*.$$

Quantitative evaluation

Method	ADD(-S)	RSR (%)	FPS
FoundationPose	0.95	0	16
Any6D	0.96	0	13
ZeroPose	0.83	0	15
Category-level	0.76	0	N/R
CLM (Ours)	**0.95**	**96 (click)**	**19**

Ablation of color–shape scale recovery

Method	ADD(-S)	RSR (0–1)
w/o color + shape	0.64	0.46
w/o shape	0.72	0.61
w/o color	0.85	0.68
Full model	**0.96**	**0.97**

Metrics: ADD(-S) recall at 0.1d. RSR follows the paper's click-guided recovery protocol. FPS excludes initialization and scale recovery. N/R = not reported.